INCH-POUND

MIL-C-50652/15C (AR)
29 March 1991
SUPERSEDING
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16 May 1990

# MILITARY SPECIFICATION SHEET

## CIRCUIT CARD ASSEMBLY: 10548330
## (CONVERTER, RANGE RATE D/A 4A17)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-50652.

REQUIREMENTS

1. Materials. Materials shall be in accordance with the applicable drawings and specifications.

2. Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548330, all associated drawings, and with all requirements specified in applicable specifications and standards. The assembly shall meet all the requirements of MIL-C-50652.

3. Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram.

4. Performance. For the requirements of 4.5 through 4.8, logic input levels shall be as follows:

Logic 0: 0.0 ±0.5 volts direct current (Vdc) level.
Logic 1: +2.8 to 5.4 Vdc level.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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4.1 Input voltages. The input voltages shall be as specified in Table I. All voltages shall be regulated with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

4.2 Bias voltage. The bias voltage measured at TP4 of the range rate D/A converter assembly shall be within the limits of +2.0 and +2.2 Vdc.

4.3 -310M/sec reference. The reference voltage measured at P1-12 and P1-14 shall be adjustable by R4 between the limits of -9.90 and -10.10 Vdc.

4.4 +310M/sec reference. The reference voltage measured at P1-16 shall be within the limits of +9.90 and +10.10 Vdc.

TABLE I. Input voltages

| VOLTAGE | TERMINALS |
|---|---|
| +5.0 ±0.3 Vdc | P1-30, P1-31, P1-32 |
| +5.0 V return | P1-24, P1-25, P1-26 |
| +15.0 ±0.3 Vdc | P1-36, P1-37 |
| ±15.0 V return | P1-22, P1-23, P1-34, P1-35 |
| -15.0 ±0.3 Vdc | P1-20, P1-21 |

4.5 Positive gain adjust. With the logic levels applied as specified in step 1 of Table II, the output voltage at P1-41 shall be adjustable by R15 between the limits as specified in step 1 of Table II.

4.6 Negative gain adjust. With the logic levels applied as specified in step 2 of Table II, the output voltage at P1-41 shall be adjustable by R16 between the limits as specified in step 2 of Table II.

4.7 Range rate voltages. With the logic levels applied sequentially for steps 3 through 12 as specified in Table II the range rate output voltages at P1-41 shall be within the limits as specified in Table II.

4.8 Range rate ratios. With the logic levels applied sequentially as specified in Table III, the ratio of the range rate output voltage at P1-41 to the reference output voltage terminal specified in Table III, shall be within the limits as specified in Table III.



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TABLE II. Range rate voltages

| STEP | EQUIVALENT1/ RANGE RATE METERS/SEC(M/S) | INPUT LOGIC LEVELS (REFERENCE P1 TERMINALS) 52 | 2 | 6 | 4 | 44 | 46 | 42 | 51 | OUTPUT VOLTAGE (REFERENCE TERMINAL P1-41) NOT LESS THAN | NOT MORE THAN |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | +310 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | +9.900 | +10.100 |
| 2 | -310 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | -9.900 | -10.100 |
| 3 | +160 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | +4.990 | + 5.090 |
| 4 | + 80 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | +2.470 | + 2.570 |
| 5 | + 40 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | +1.210 | + 1.310 |
| 6 | + 20 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | +0.580 | + 0.680 |
| 7 | + 10 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | +0.265 | + 0.365 |
| 8 | + 5 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | +0.107 | + 0.207 |
| 9 | + 2.5 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | +0.028 | + 0.128 |
| 10 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -0.050 | + 0.050 |
| 11 | -160 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | -4.910 | - 5.010 |
| 12 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | -0.050 | + 0.050 |

1/ Range rates are listed by their schematic value, a rounded value, except for the +310 and -310 which is the true maximum range rate for which this board is adjusted.



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TABLE III. Range rate ratios

| STEP | EQUIVALENT RANGE RATE METERS/SEC (M/S) | INPUT LOGIC LEVELS (REFERENCE P1 TEMINALS 52 | 2 | 6 | 4 | 44 | 46 | 42 | 51 | RANGE RATE RATIO NOT LESS THAN | NOT MORE THAN | REF TERM P1 | RATIO MODE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | +310 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | +0.9802 | +1.0202 | 16 | + |
| 2 | -310 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | -0.9802 | -1.0202 | 14 | - |
| 3 | +160 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | +0.4940 | +0.5141 | 16 | + |
| 4 | +80 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | +0.2446 | +0.2596 | 16 | + |
| 5 | +40 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | +0.1198 | +0.1323 | 16 | + |
| 6 | +20 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | +0.0574 | +0.0687 | 16 | + |
| 7 | +10 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | +0.0263 | +0.0369 | 16 | + |
| 8 | +5 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | +0.0106 | +0.0209 | 16 | + |
| 9 | +2.5 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | +0.0028 | +0.0129 | 16 | + |
| 10 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | -0.0050 | +0.0050 | 16 | + |
| 11 | -160 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | -0.5141 | -0.4940 | 14 | - |
| 12 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | -0.0050 | +0.0050 | 14 | - |



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5. Test Equipment. The test equipment listed in Table IV shall be used in the performance of the applicable test methods and procedures as specified in 6.

TABLE IV. Test equipment list

| ITEM NUMBER | DESCRIPTION |
|---|---|
| 1. Power supply | 5.0 ±0.3 Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity, 200 milliamps (mA). |
| 2. Power supply | 15.0 ±0.3 Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity, 200 mA |
| 3. Power supply | Same as item 2. |
| 4. Digital volt-ratio meter | Visual display: 5 digits<br>Dc range: 0.0000 to ±999.99 V.<br>Accuracy ±10.01% of reading +0.0001% of full scale.<br>Dc input resistance: Greater than 10 megohms.<br>Dc ratio: 1.0999 to 1, ±0.005% of reading +0.002% of full scale with ±10 V minimum reference.<br>Common mode rejection: Dc greater than 134 dB, 60 Hz greater than 120 dB.<br>Response time: 1500 milliseconds (ms) maximum. |
| 5. Test fixture | As specified in Figure 1. See Table V for test fixture parts list. |



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TABLE V. Test fixture parts list

| DESIGNATION | NOMENCLATURE AND DESCRIPTION |
|---|---|
| E1 thru E4 | Binding post: H.H. Smith 257 or equivalent. |
| J1 | Connector: any of the following or equivalent.<br>(1) 5893232, Malco Mfg. Co., Inc., Chicago, Illinois<br>(2) 500946-01, National Connector Corp. Minneapolis, Minn.<br>(3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa. |
| S1 thru S8 | Switch SPDT |
| TP1 thru TP5 | Tip jack: H.H. Smith 1501 or equivalent |

6 Methods of inspection.

6.1 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with requirement 3.

6.2 Performance. The range rate D/A converter assembly shall be tested as follows: Connect range rate D/A converter and test equipment as shown in Figure 2. Switches S1 through S8, on the test fixture (Table IV, item 7) are used to supply the logic level input signals as specified in Tables II and III. These switches supply a logic 1 in position 1 and a logic 0 in position 2.

6.2.1 Input voltages. Using power supplies (Table IV, items 1 through 3), apply input voltages as specified in Table I.

6.2.2 Bias voltage. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the bias voltage TP4 of the range rate D/A converter to be as specified in requirement 4.2. Use TP5 on the test fixture (Table IV, item 5) as ground reference.



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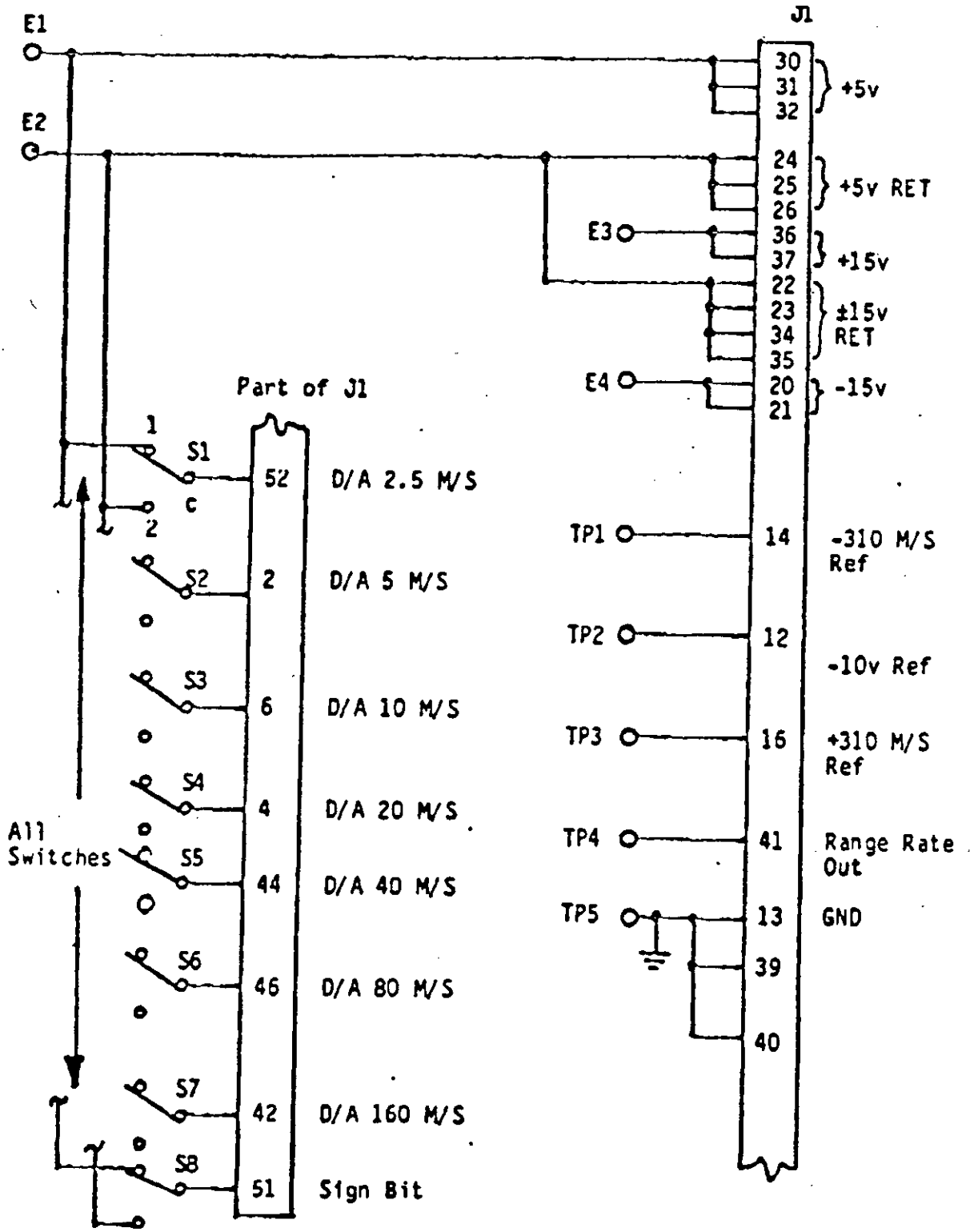
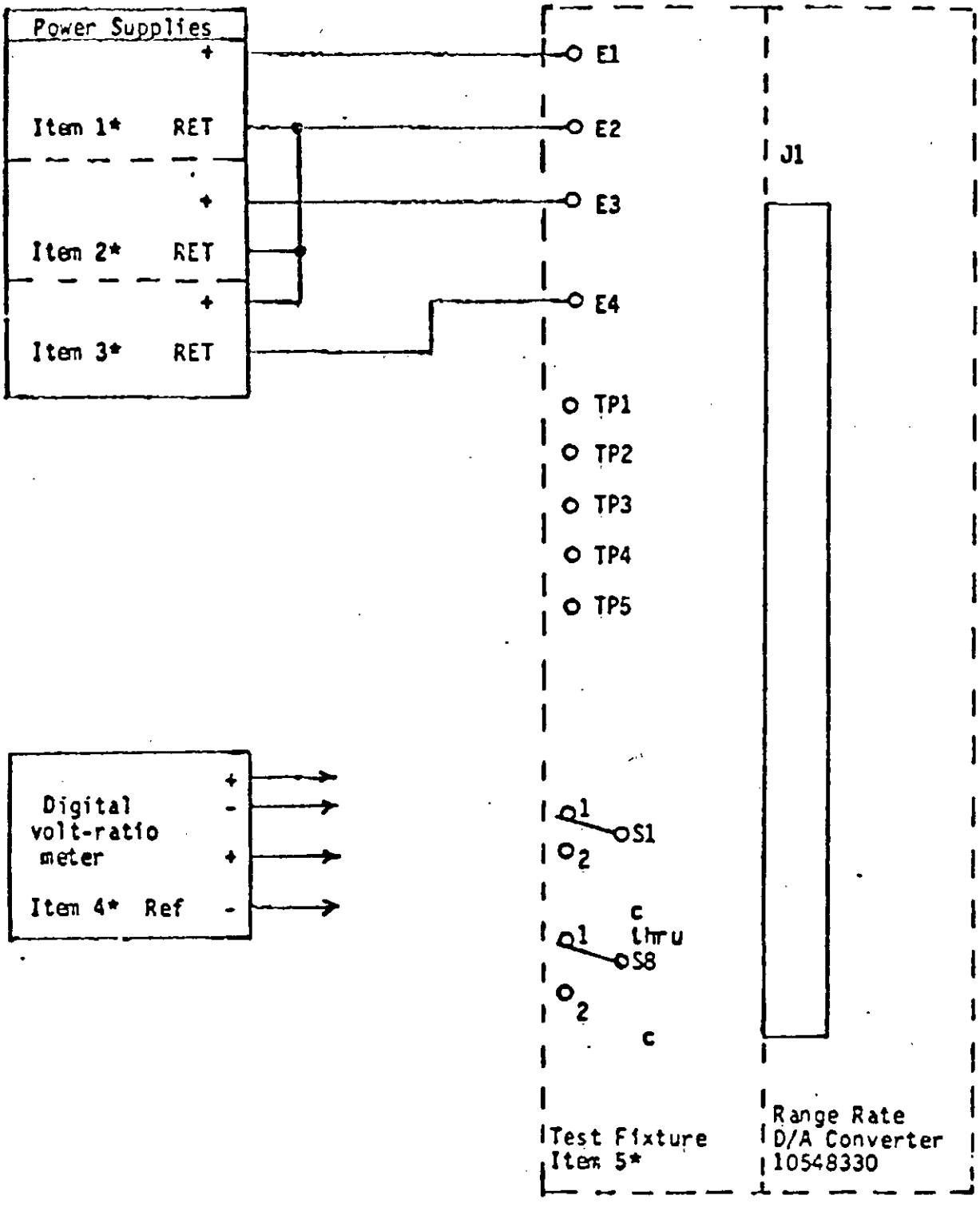



FIGURE 1. Test Fixture setup.



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*Refer to 5

FIGURE 2. Range rate D/A converter test setup.



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6.2.3 -310 M/sec reference. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the -310 volt reference at TP1 and TP2 on the test fixture to be as specified in requirement 4.3. Use TP5 on the test fixture (Table IV, item 5) as ground reference.

6.2.4 +310 M/sec reference. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the +310 volt reference at TP3 on the test fixture to be as specified in requirement 4.4. Use TP5 on the test fixture (Table IV, item 5) as ground reference.

6.2.5 Positive gain adjust. Using switches S1 and S8 on the test fixture (Table IV, item 5) apply the logic levels as specified in step 1 of Table II. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the positive gain adjust at TP4 on the test fixture to be as specified in requirement 4.5. Use TP5 on the test fixture as ground reference.

6.2.6 Negative gain adjust. Using switches S1 through S8 on the test fixture (Table IV, item 5) apply the logic levels as specified in step 2 of Table II. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the negative gain adjust at TP4 on the test fixture to be as specified in requirement 4.6. Use TP5 on the test fixture as ground reference.

6.2.7 Range rate voltages. Using switches S1 through S8 on the test fixture (Table IV, item 5) sequentially apply the logic levels as specified in steps 3 through 12 of Table II. Using the volt-ratio meter (Table IV, item 4) in the voltage mode, measure the range voltages at TP4 on the test fixture to be as specified in requirement 4.7. Use TP5 on the test fixture as ground reference.

6.2.8 Range rate ratios. Using switches S1 through S8 on the test fixture (Table IV, item 5) sequentially apply the logic levels as specified in the steps of Table III. Using the volt-ratio meter (Table IV, item 4) in the ratio mode, measure the range rate ratios to be as specified in requirement 4.8. Connect the volt-ratio meter leads to the test fixture test points as follows:

| | | |
|---|---|---|
| + voltage lead | - TP4 | |
| - voltage lead | - TP5 | |
| + reference lead | - TP1 | for P1-14 (- ratio mode) |
| | TP3 | for P1-16 (+ ratio mode) |
| - reference lead | - TP5 | |

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